ModelIt!

Teacher's Guide

Growing Meat Without the Animal

Optimizing Cultured Meat Bioreactor Systems from Cell to Steak

Quick Reference

Grade Level	Grade 09L3
NGSS Standard	HS-LS1-2, HS-ETS1-3
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS1-2, HS-ETS1-3

Develop and use a model to illustrate the hierarchical organization of interacting systems that provide specific functions within multicellular organisms; evaluate a solution to a complex real-world problem based on prioritized criteria and trade-offs.

Three-Dimensional Learning

Science Practice	**Developing and Using Models / Constructing Explanations and Designing Solutions** — Students develop a computational bioreactor model and use it to design optimized solutions that balance competing production constraints.
Disciplinary Core Idea	**LS1.A Structure and Function / ETS1.C Optimizing the Design Solution** — Multicellular organisms have hierarchical organization; engineering optimization requires systematically testing and refining solutions against multiple criteria and constraints.
Crosscutting Concept	**Systems and System Models / Stability and Change** — Students model the bioreactor as a complex system where multiple interacting variables must be simultaneously optimized, and small changes in one parameter cascade through the entire system.

CAST Testing Alignment

MULTIPLE CHOICE:

A cultured meat company scales their bioreactor from 10 liters to 10,000 liters. Which problem is most likely to worsen at larger scale, based on your model?

CONSTRUCTED RESPONSE:

Using your bioreactor optimization model, explain why cultured meat that tastes identical to conventional beef is so difficult to produce at affordable prices. Reference at least three model components and the trade-offs between them.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

This lesson introduces students to predictive modeling and multi-parameter optimization in the context of cellular agriculture — one of the most ambitious bioengineering challenges of the 21st century. Biotech skill focus: Predictive modeling and optimization. Growing meat from cell cultures requires simultaneously optimizing biological, engineering, and economic parameters — a perfect application for computational modeling.

Cellular Agriculture Fundamentals

Cultured meat begins with a small biopsy of muscle stem cells (satellite cells) from a living animal — a painless procedure that yields enough cells to theoretically produce tons of meat. These cells are placed in nutrient-rich media inside bioreactors where they proliferate, differentiate into muscle fibers, and organize into tissue. The process mirrors what happens naturally in an animal's body: cells divide, consume nutrients, produce waste, and organize into structured tissues. The difference is that engineers must artificially provide everything the animal's body normally supplies — nutrients, oxygen, waste removal, structural support, growth signals, and mechanical stimulation. The current state of the art can produce small quantities of cultured meat that is safe to eat and reasonably similar to conventional meat, but at costs that are 10-40x higher than conventional production.

The Bioreactor Engineering Challenge

Bioreactors for cultured meat must maintain mammalian cell culture conditions (37°C, pH 7.4, 20% dissolved oxygen, specific nutrient concentrations) while scaling to thousands of liters. The critical engineering challenge is mass transfer — getting oxygen and nutrients TO cells and waste products AWAY from cells throughout the entire volume. In a small flask, diffusion handles this naturally. In a 10,000-liter bioreactor, cells in the center can be centimeters from the nearest oxygen source, far beyond the ~200 micrometer diffusion limit for mammalian cells. Solutions include perfusion bioreactors (continuous medium flow), hollow-fiber systems (oxygen delivered through permeable tubes), and microcarrier cultures (cells growing on small beads suspended in circulating medium). Each approach has trade-offs in cost, scalability, and product quality.

The Growth Factor Cost Problem

The single biggest cost barrier in cultured meat is the growth medium — specifically, the growth factors (signaling proteins like FGF2, TGF-beta, IGF-1, and transferrin) that instruct cells to proliferate and differentiate. In research labs, these are purchased at costs of $500-$50,000 per gram. Traditional cell culture uses fetal bovine serum (FBS) as a growth factor source, but FBS costs $300-500/liter, is derived from unborn calves (defeating the ethical purpose), and varies batch-to-batch. The industry is racing to develop serum-free, animal-free media using recombinant growth factors produced by engineered bacteria or yeast, edible plant-derived alternatives, or genetic engineering of the meat cells themselves to produce their own growth factors (autocrine signaling). Reducing medium cost from ~$400/liter to under $1/liter is necessary for commercial viability.

Texture: The Final Frontier

The biggest quality challenge in cultured meat is texture. Ground meat products (burgers, nuggets, sausages) are relatively straightforward because they don't require organized tissue structure. But whole-cut products (steaks, chicken breasts, pork chops) require aligned muscle fibers, intramuscular fat marbling, connective tissue, and the complex architecture that gives meat its 'bite.' Achieving this requires three-dimensional scaffolds (made from plant proteins, alginate, or decellularized plant tissue), mechanical stimulation (cyclic stretching that mimics exercise), and co-culture of multiple cell types (muscle, fat, fibroblasts). Current prototypes achieve reasonable texture in thin products (<5mm thickness) but struggle with thick cuts where oxygen delivery and structural integrity become limiting factors.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify ten components spanning cell biology (Cell Growth Rate, Waste Product Accumulation), engineering (Bioreactor Temperature, Oxygen Supply, Scaffold Structure, Energy Input), medium chemistry (Nutrient Medium Concentration), and economics (Production Cost, Scale Factor, Texture Quality).

E **Establish Relationships**

Students determine that Nutrient Medium Concentration, Bioreactor Temperature, Oxygen Supply, Scaffold Structure, and Energy Input are the controllable inputs that drive Cell Growth Rate and Texture Quality — while Waste Product Accumulation acts as a growth inhibitor, and Production Cost and Scale Factor represent the economic constraints.

V **Visualize & Model**

Students build a computational model connecting all ten components, visualizing how changing one parameter cascades through the system and affects multiple outputs simultaneously.

E **Evaluate Outcomes**

Students run lab-scale, maximum-growth, and commercial-scale scenarios to discover that optimizing for one metric (speed, quality, or cost) often degrades the others — and search for the narrow window where all three are acceptable.

R **Revise & Extend**

Students add Fat Cell Integration, Mechanical Stimulation, or Serum-Free Medium Development to explore more realistic cultured meat production challenges.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"By 2050, the world needs to feed 10 billion people. Conventional livestock uses 77% of agricultural land but produces only 18% of calories. What if we could grow the same meat — same taste, same texture — without a single animal?"*

Watch For

Show the land use statistic with a visual comparison. Ask students to react before introducing cultured meat.

Slide 2: Learning Objectives

3 min

Say: *"Today you're bioprocess engineers. Your job is to optimize a bioreactor that grows real beef from cells — and figure out how to make it affordable enough for everyone."*

Watch For

Pre-teach bioreactor and scaffold. Show a 30-second time-lapse of cells growing in culture to establish the biological basis.

Slide 3: Big Question

3 min

Say: *"The first cultured beef burger cost $330,000 to produce in 2013. Today it's down to around $10-50 per burger. But conventional beef burgers cost $1. How do we close that gap — and what has to be sacrificed along the way?"*

Watch For

Quick-write: Students list what they think makes cultured meat expensive. Compile class predictions on the board.

Slide 4: LEVER Framework

2 min

Say: *"We're going to optimize ten interacting parameters simultaneously — cell biology, engineering, and economics all at once. Welcome to multi-parameter optimization."*

Watch For

Preview LEVER steps. Emphasize that this model has five external (controllable) variables — the most of any model we've built.

Slide 5: Activity 1: Components

8 min

Say: *"Five of these ten components are controllable — you set them. The other five are responses. Identify them and think about why the controllable ones are the expensive ones."*

Watch For

Guide sorting of external versus internal components. Note that four of five externals represent costs (nutrients, energy, temperature control, scaffolds) — this is why cultured meat is expensive.

Slide 6: Activity 2: Connections

8 min

Say: *"Here's the optimization nightmare: making cells grow faster increases waste. Improving texture adds cost. Scaling up breaks oxygen delivery. Everything is connected to everything."*

Watch For

Students map relationships, identifying trade-offs and constraints. Highlight the fundamental tension between quality, speed, and cost.

Slide 7: Activity 3: Simulation

10 min

Say: *"Find the sweet spot. The bioreactor configuration where cells grow fast enough, meat tastes good enough, and the price is low enough. Spoiler: it's a very small target."*

Watch For

Students predict outcomes, then run lab-scale, max-growth, and commercial-scale scenarios. Track the three-way trade-off between growth rate, texture quality, and production cost.

Slide 8: Discoveries

5 min

Say: *"The industry has already brought costs down 99.9% from $330,000 to about $10 per burger. Your model shows the same optimization principles the real companies are using."*

Watch For

Connect model findings to the real cost trajectory of cultured meat. Discuss which remaining bottlenecks your model identified.

Slide 9: STEM Challenge

12 min

Say: *"A startup just raised $50 million to build the world's first commercial cultured meat factory. Design their bioreactor."*

Watch For

Groups design commercial bioreactor systems including vessel type, monitoring systems, waste management, and cost projections. Present engineering proposals.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Nutrient Medium Concentration, Bioreactor Temperature, Oxygen Supply, Scaffold Structure, Energy Input
- **INTERNAL:** Cell Growth Rate, Waste Product Accumulation, Texture Quality, Production Cost, Scale Factor

Relationships

- **Nutrient Medium Concentration to Cell Growth Rate:** POSITIVE (+) — Higher nutrient concentration provides more amino acids, glucose, and growth factors for cell metabolism and division, directly increasing the cell proliferation rate.
- **Cell Growth Rate to Waste Product Accumulation:** POSITIVE (+) — Faster-growing cells metabolize more nutrients per unit time, producing proportionally more waste products (ammonia, lactate) that accumulate in the medium.
- **Waste Product Accumulation to Cell Growth Rate:** NEGATIVE (-) — As metabolic waste accumulates, it creates an increasingly toxic environment that inhibits cell division — this is the primary negative feedback loop in bioreactor systems.
- **Oxygen Supply to Cell Growth Rate:** POSITIVE (+) — Adequate dissolved oxygen is essential for aerobic cellular metabolism; insufficient oxygen causes cells to switch to inefficient anaerobic pathways or die from hypoxia.
- **Scaffold Structure to Texture Quality:** POSITIVE (+) — Better scaffold design provides more appropriate physical cues for cell alignment, fiber formation, and tissue architecture, directly improving the textural properties of the final product.
- **Nutrient Medium Concentration to Production Cost:** POSITIVE (+) — Growth medium — especially growth factors — is the primary cost driver; higher concentration and more frequent replacement directly increase per-kilogram production costs.

Key Discoveries

- The primary cost driver in cultured meat is the nutrient medium — particularly growth factors that currently cost thousands of dollars per gram and must be produced through recombinant protein technology
- Oxygen delivery is the critical scaling bottleneck — cells more than 200 micrometers from an oxygen source become hypoxic, limiting how thick cultured meat tissue can grow without vascularization
- Waste product accumulation creates a negative feedback loop — as cells grow and metabolize, they produce toxins that inhibit further growth, requiring continuous medium perfusion or replacement
- Texture quality requires both scaffold engineering and mechanical stimulation — without aligned fibers and exercised muscle cells, cultured meat lacks the sensory experience consumers expect from conventional meat

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Lab-grown meat is 'fake' or 'artificial' meat"	Cultured meat is real animal meat — it contains the same muscle cells, proteins, and fats as conventional meat because it IS animal cells. The only difference is where the cells grew: in a bioreactor instead of inside an animal. At the molecular level, a cultured beef cell is identical to a cell from a cow's muscle. Calling it 'fake' is like calling ice from a freezer 'fake ice' because it wasn't made by a glacier.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Cultured meat must be unhealthy because it's 'unnatural'"	Cultured meat can be engineered to be HEALTHIER than conventional meat. Producers can control fat content, reduce saturated fat, add omega-3 fatty acids, eliminate antibiotics and growth hormones, and ensure zero pathogen contamination (no Salmonella, E. coli, or prions). The 'unnatural' criticism ignores that most modern food production — from cheese to bread to yogurt — involves controlled biological processes.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Cultured meat will be available and affordable within a few years"	While progress has been extraordinary ($330,000/burger in 2013 to ~$10-50 today), achieving true commercial viability ($5-15/kg at millions of tons/year) requires solving several fundamental engineering challenges that the model reveals: growth factor cost reduction by 100-1000x, oxygen delivery at industrial scale, waste management for massive bioreactors, and texture engineering for whole-cut products. Most industry experts estimate 5-15 years for cost-competitive commodity products, though premium products may arrive sooner.